

Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -2(Adv-2025-P2-Model)** **Date: 05-04-2026**

Time: 02.30Pm to 05.30Pm **GTA-22** **Max. Marks: 180**

05-04-2026_Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT_Jee-Adv(2025-P2)_GTA-22_Syllabus,

PHYSICS : TOTAL SYLLABUS

CHEMISTRY : TOTAL SYLLABUS

MATHEMATICS : TOTAL SYLLABUS

Rankers Academy JEE
Sri Chaitanya Educational Institutions INDIA

Name of the Student: _____ **H.T. NO:** | | | | | | | | |

Sri Chaitanya Educational Institutions
Infinity Learn
41


For More Join: @IITJEE_Advanced

JEE-ADVANCED-2025-P2-Model

Time:3Hr's **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 4)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 5 – 8)	Questions with Multiple Correct Choice with partial mark	+4	-2	4	16
Sec – III(Q.N : 8 – 13)	NUMERICAL VALUE	+4	0	8	32
Total				**16**	**60**

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 17 – 20)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 21 – 24)	Questions with Multiple Correct Choice with partial mark	+4	-2	4	16
Sec – III(Q.N : 25 – 32)	NUMERICAL VALUE	+4	0	8	32
Total				**17**	**60**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 33 – 36)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 37 – 40)	Questions with Multiple Correct Choice with partial mark	+4	-2	4	16
Sec – III(Q.N : 41 – 48)	NUMERICAL VALUE	+4	0	8	32
Total				**17**	**60**

Sri Chaitanya Educational Institutions
Infinity Learn by Sri Chaitanya
41 Years of Academic Excellence 1986-2026
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 RANK 1 SINGARAJU VENKAT KOUNDINYA sri chaitanya 6th-12th Class
JEE Advanced 2023 RANK 1 VAVILALA CHIDVILAS REDDY sri chaitanya 6th-12th Class
NEET 2023 RANK 1 BORA VARUN CHAKRAVARTHI sri chaitanya 6th-12th Class
For More Join: @IITJEE_Advanced

MATHEMATICS **Max Marks: 60**

SECTION–I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

1. The mirror image of the curve $\arg\left(\frac{z+i}{z-1}\right)=\frac{\pi}{4}$ where $i=\sqrt{-1}$ in the line $y=x$ is

A) $\arg\left(\frac{z+i}{z+1}\right)=\frac{\pi}{4}$ **B)** $\arg\left(\frac{z+1}{z-i}\right)=\frac{\pi}{4}$ **C)** $\arg\left(\frac{z-i}{z+1}\right)=\frac{\pi}{4}$ **D)** $\arg\left(\frac{z+i}{z-1}\right)=\frac{\pi}{4}$

2. A circle of constant radius r passes through the origin 0 and cuts the axes at A and B. The locus of the foot of the perpendicular from 0 to AB is

$\left(x^a+y^b\right)^c\left(x^{-\alpha}+y^{-\beta}\right)=\lambda r^8$ then the value of $(a+b+c+\alpha+\beta+\lambda)$ is

A) 10 **B)** 12 **C)** 14 **D)** 16

3. If $\sin\theta\sec(3\theta)+\sin 3\theta.\sec\left(3^2\theta\right)+\sin\left(3^2\theta\right)\sec\left(3^3\theta\right)+.....$up to 10 terms when $\theta=\frac{\pi}{3^5}$ is

A) $-\frac{1}{2}\tan\left(\frac{\pi}{3^5}\right)$ **B)** $\frac{1}{2}\tan\left(\frac{\pi}{3^5}\right)$ **C)** $\frac{1}{2}\tan\left(\frac{\pi}{3^{10}}\right)$ **D)** $-\frac{1}{2}\tan\left(\frac{\pi}{3^{10}}\right)$

4. If $2x=y^{-1/5}+y^{1/5}$ *and* $f(x).\frac{d^2y}{dx^2}+g(x).\frac{dy}{dx}=\lambda y$ then the number of real roots of $f(x)+g(x)+\lambda=0$ is

A) 4 **B)** 3 **C)** 2 **D)** 0

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.
For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

Sri Chaitanya Educational Institutions
Infinity Learn


For More Join: @IITJEE_Advanced

5. If $\int \frac{\tan^{-1}x}{x^4}dx = -\frac{1}{3x^3}\left(f(x)\right)+\frac{1}{6}\log\left|g(x)\right|+h(x)+c$ then

A) $f(1)=\frac{\pi}{12}$ **B)** $g(2)=\frac{4}{5}$ **C)** $h\left(\frac{2}{3}\right)=-\frac{3}{8}$ **D)** $h\left(\frac{2}{3}\right)=\frac{3}{8}$

6. If $\alpha=\sum_{k=0}^{10}\frac{\left({}^{10}C_k\right)^2}{(k+1)}$ *and* $\beta=\sum_{k=0}^{9}\frac{\left({}^{10}C_k.{}^{10}C_{k+1}\right)}{(k+2)}$ then

A) $\alpha=\frac{{}^{21}C_{11}}{11}$ **B)** $\alpha=\frac{{}^{20}C_{10}}{10}$ **C)** $\beta=\frac{{}^{20}C_{12}}{10}$ **D)** $\frac{\alpha}{\beta}=\frac{6}{5}$

7. Let P be a square matrix such that $P^2=I-P$. For $\alpha,\beta,\gamma,\delta\in N$, if $P^\alpha+P^\beta=\gamma I+29P$, $P^\alpha-P^\beta=6I-13P$ then

A) $\alpha+\beta=14$ **B)** $\beta+\gamma=24$ **C)** $\gamma+\delta=26$ **D)** $\alpha+\beta+\gamma-8=24$

8. Let the shortest distance between the lines

$L:\frac{x-5}{-2}=\frac{y-\lambda}{0}=\frac{z-\lambda}{1},\lambda\geq 0$ *and* $L_1: x+1=y-1=4-z$ be $\sqrt{26}$. If (α,β,γ) lies on L, then which of the following is possible?

A) $\alpha+2\gamma=24$ **B)** $2\alpha+\gamma=7$ **C)** $\alpha-2\gamma=19$ **D)** $2\alpha-\gamma=9$

SECTION-III
(NUMERICAL VALUE)

- This section contains **EIGHT (08)** questions.
- The answer to each question is a **NUMERICAL VALUE**..
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4If ONLY the correct integer is entered;
Zero Marks : 0 In all other cases.

9. *let* $f(x)=\begin{cases}\max\left\{|x|,x^2\right\};|x|\leq 2\\ 8-2|x|;2<|x|\leq 4\end{cases}$ Let S be the set of points in the interval $(-4,4)$ at which f is not differentiable. Then sum of the elements of S is

Sri Chaitanya Educational Institutions
Infinity Learn
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 SINGARAJU VENKAT KOUNDINYA RANK 1
JEE Advanced 2023 VAVILALA CHIDVILAS REDDY RANK 1
NEET 2023 BORA VARUN CHAKRAVARTHI RANK 1
For More Join: @IITJEE_Advanced

10. Let $A=\left[a_{ij}\right], a_{ij}\in Z\cap[0,4], 1\le i,j\le 2$ The number of matrices A such that the sum of all the elements is a prime number $P\in[5,12]$ is _____

11. If the area bounded by the curve $2y^2=3x$, lines $x+y=3$, $y=0$ and outside the circle $(x-3)^2+y^2=2$ is A then $10(\pi+4A)$ is equal to

12. Let the focal chord of the parabola $P: y^2=4x$ along the line $L: y=mx+c$, $m>0$ meet the parabola at the points M and N. let the line L be a tangent to the hyperbola $H: x^2-y^2=4$. If O is the vertex of P and F is the focus of H on the positive x-axis and area of the quadrilateral OMFN is Δ then $\left(\sqrt{14}\right)\Delta$ is

13. If the solution curve of the differential equation $\frac{dy}{dx}=\frac{x+y-2}{x-y}$ passing through the point $(2,1)$ is $\beta\tan^{-1}\left(\frac{y-1}{x-1}\right)-10\log\left(\frac{10}{\alpha}+\left(\frac{y-1}{x-1}\right)^2\right)=\beta\log_e|x-1|$ then the value of $5\beta+7\alpha$ is

14. If the plane $\pi: 7x+(\alpha+4)y+4z-\gamma=0$ passing through the point of intersection of the planes $2x+3y-z+1=0$ and $x+y-2z+3=0$ and is perpendicular to the plane $3x-y-2z=4$ and $\left(\frac{12}{\beta},\frac{-78}{\beta},\frac{57}{\beta}\right)$ is image of the point $(1,1,1)$ in π then $(\alpha+\beta-\gamma)$ is

15. For $K\in N, let\ \frac{1}{\alpha(\alpha+1)(\alpha+2).....(\alpha+20)}=\sum_{k=0}^{20}\frac{A_k}{a+k}$ where a>0. Then the value of $10^3\left(\frac{A_{14}+A_{15}}{A_{13}}\right)^2$ is

16. Let $L=\lim_{x\to 0}\frac{x^4+3\left(a^2-\sqrt{x^4+a^4}\right)}{x^8}, a>0$, If L is the finite, then 72L is equal to

Sri Chaitanya Educational Institutions
Infinity Learn
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 SINGARAJU VENKAT KOUNDINYA RANK 1
JEE Advanced 2023 VAVILALA CHIDVILAS REDDY RANK 1
NEET 2023 BORA VARUN CHAKRAVARTHI RANK 1
For More Join: @IITJEE_Advanced

PHYSICS **Max Marks: 60**

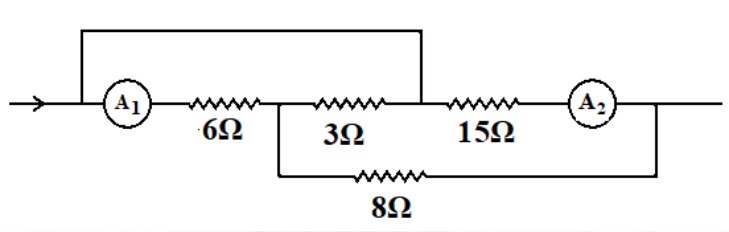
SECTION–I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

17. In the circuit shown, ammeters A_1 *and* A_2 are ideal. If reading of A_1 is $2A$, then reading of A_2 is

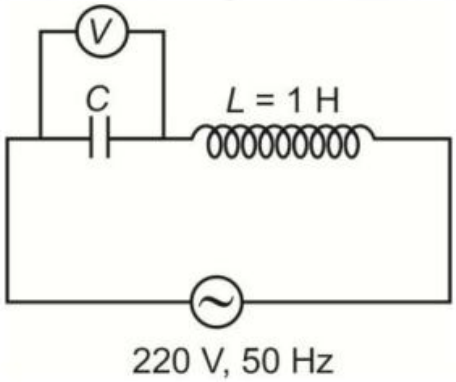


A) 6A **B)** 1A **C)** 4A **D)** 3A

18. An artificial satellite of mass *m* is orbiting around earth at low altitude. The satellite experiences a constant retarding force *F* due to atmosphere. The rate of decrease of height of satellite at the instant when velocity is *v*, is

A) $\frac{Fv}{mg}$ **B)** $\frac{2Fv}{mg}$ **C)** $\frac{Fv}{2mg}$ **D)** ZERO

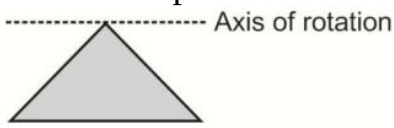
19. In the circuit shown, the ideal voltmeter reads 220 V, then value of capacitance C, is $\left(Take\,\pi^2 = 10\right)$



A) $10\mu F$ **B)** $20\mu F$ **C)** $30\mu F$ **D)** $40\mu F$

Sri Chaitanya Educational Institutions
Infinity Learn
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 SINGARAJU VENKAT KOUNDINYA RANK 1
JEE Advanced 2023 VAVILALA CHIDVILAS REDDY RANK 1
NEET 2023 BORA VARUN CHAKRAVARTHI RANK 1
For More Join: @IITJEE_Advanced

20. The moment of inertia of an equilateral triangular lamina of mass M and side length L about an axis in the plane of lamina and parallel to one side as shown in figure is



A) $\frac{5ML^2}{16}$ **B)** $\frac{ML^2}{16}$ **C)** $\frac{3}{8}ML^2$ **D)** $\frac{ML^2}{24}$

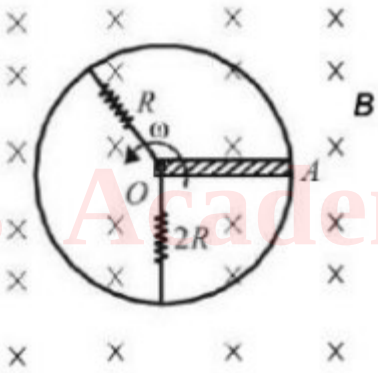
SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.
For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks;

21. A rod OA of length ℓ is rotating (about end O) over a conducting ring in crossed magnetic field B with constant angular velocity ω as shown in figure



A) Current flowing through the rod is $\frac{3B\omega\ell^2}{4R}$

B) Magnetic force acting on the rod is $\frac{3B^2\omega\ell^2}{4R}$

C) Torque due to magnetic force acting on the rod is $\frac{3B^2\omega\ell^4}{8R}$

D) Magnitude of external force that acts perpendicularly at the end of the rod to maintain the constant angular speed is $\frac{3B^2\omega\ell^4}{8R}$

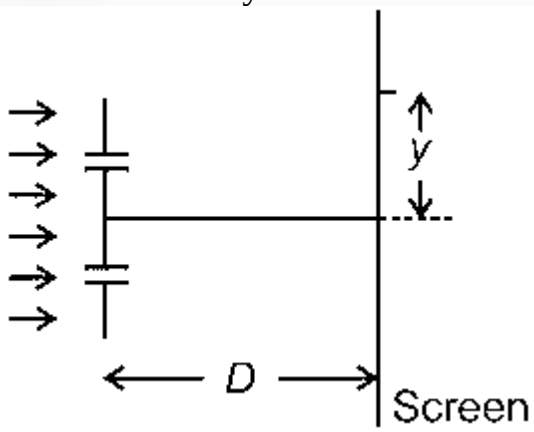
Sri Chaitanya Educational Institutions
Infinity Learn
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 SINGARAJU VENKAT KOUNDINYA RANK 1
JEE Advanced 2023 VAVILALA CHIDVILAS REDDY RANK 1
NEET 2023 BORA VARUN CHAKRAVARTHI RANK 1
For More Join: @IITJEE_Advanced

22. A body of mass 100 g starts at $t = 0$ with velocity of $2\,m/s$ and accumulates dust at the rate of $5\,g/s$. Then

A) Velocity of body at $t = 20$ s is 1 m/s **B)** Velocity of body at $t = 20$ s is 2 m/s

C) Distance covered in $\Delta T = 20s$ is 27.7m **D)** Distance covered in $\Delta T = 20s$ is 35.2 m

23. In a standard YDSE arrangement $\lambda = 1mm, d = 3mm$ and $D = 1m$. A plane monochromatic wave is incident normally on slits as shown. Then



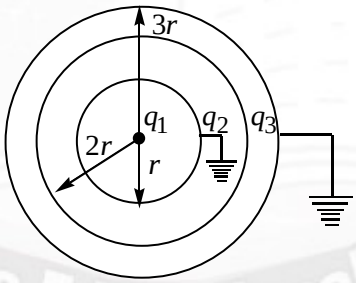
A) Distance of first maxima from central maxima is $y = \frac{1}{\sqrt{8}}m$

B) Distance of second maxima from central maxima is $y = \frac{2}{\sqrt{5}}m$

C) Number of maxima on the screen is 5 (excluding at infinity)

D) Number of minima on the screen is 6

24. Three concentric conducting spherical shells have radii r, 2r and 3r and charge q_1, q_2 *and* q_3 respectively as shown in figure. Select the correct alternatives



A) $q_1 + q_3 = -q_2$ B) $q_1 = -\frac{q_2}{4}$ C) $\frac{q_3}{q_1} = 3$ D) $\frac{q_3}{q_2} = -\frac{1}{3}$

SECTION-III
(NUMERICAL VALUE)

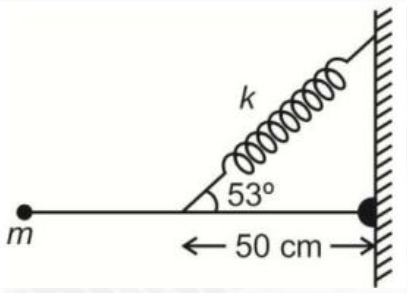
- This section contains **EIGHT (08)** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Sri Chaitanya Educational Institutions
Infinity Learn
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 RANK 1
JEE Advanced 2023 RANK 1
NEET 2023 RANK 1
For More Join: @IITJEE_Advanced

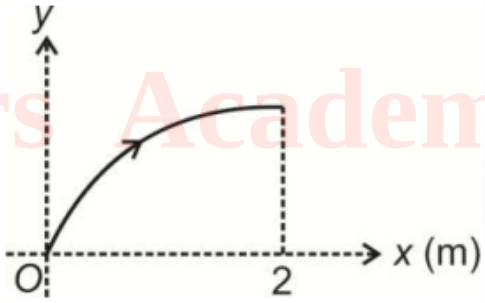
Full Marks: +4If ONLY the correct integer is entered;
Zero Marks : 0 In all other cases.

25. A uniform wire of mass m, density d and resistivity ρ is in the form of planner loop. A uniform time varying magnetic field $B = \alpha t$ is present perpendicular to plane of the loop. The maximum current through the loop is given by $I_{max} = k \times \frac{\alpha m}{\pi \rho d}$. Find value of K

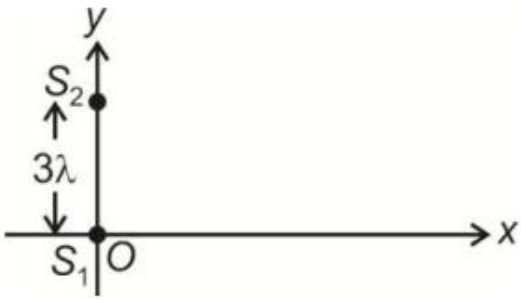
26. A small ball of mass $m = 1kg$ is attached at one end of light rod of length $1m$ and other end of rod is hinged to a support as shown. A spring of $k = 100 N/m$ is also attached to the rod as shown. The entire arrangement is on horizontal smooth surface. If the ball is displaced vertically slightly and released, then find the time period of oscillations in seconds.



27. A conducting wire section is bent in the form of a parabola $y^2 = 2x$ as shown in figure. It carries a current of $5A$. A uniform magnetic field $\vec{B} = 4\hat{k}T$ exists in the region. Find the magnetic force on the wire (in Newton).

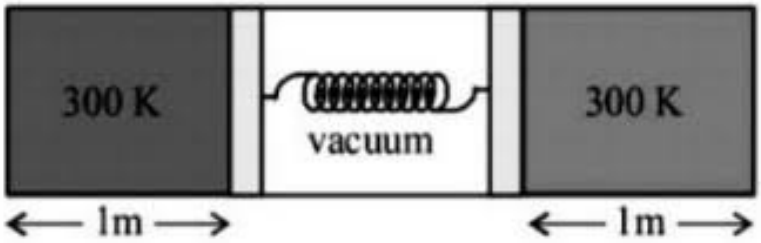


28. Two coherent point sources S_1 *and* S_2 are placed at separation of 3λ as shown in figure. A detector is moved along positive x-direction starting from origin. The x-coordinate at which the detector registered second minima is given as $x_2 = k\lambda$. Find the value of K is



Sri Chaitanya Educational Institutions
Infinity Learn
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 SINGARAJU VENKAT KOUNDINYA RANK 1
JEE Advanced 2023 VAVILALA CHIDVILAS REDDY RANK 1
NEET 2023 BORA VARUN CHAKRAVARTHI RANK 1
For More Join: @IITJEE_Advanced

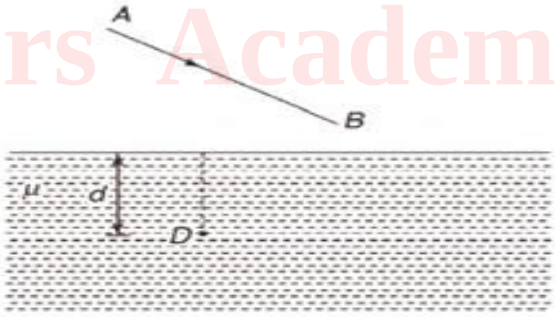
29. Consider the shown diagram where the two chambers separated by piston-spring arrangement contain equal amounts of certain ideal gas. Initially when the temperatures of the gas in both the chambers are kept at 300K the compression in the spring is 1m. The temperature of the left and the right chambers are now raised to 400K and 500 K respectively. if the pistons are free to slide, the compression in the spring is about. (Round off your answers to 1 digit after decimal)



30. A stationary hydrogen atom emits a photon corresponding to first line of Balmer series. If mass of hydrogen atom is *m* then recoil speed of hydrogen atom is given by $\frac{1}{k}\times\left(\frac{hR}{m}\right)$.

Find value of K.

31. A diver D is still under water $\left(\mu=\frac{4}{3}\right)$ at a depth d=10m. A bird is diver along line AB at constant velocity in air. When the bird is exactly above the diver, he sees it at a height of 50m from himself and velocity of the bird appears to be inclined at 45° to the horizontal. At what distance form the diver the bird actually hits the water surface.



32. The pitch of the screw gauge is 1mm and there are 100 divisions on its circular scale. When nothing is put between the jaws the zero of circular scale lies 5 divisions below the reference line. The screw gauge is used to measure the diameter of a wire. The main scale reading is 3mm and the 40th circular divisions coincide with main scale. If diameter of wire is K mm, then the value $\frac{100}{5}(k-3)$ is_____

Sri Chaitanya Educational Institutions
Infinity Learn
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 SINGARAJU VENKAT KOUNDINYA RANK 1
JEE Advanced 2023 VAVILALA CHIDVILAS REDDY RANK 1
NEET 2023 BORA VARUN CHAKRAVARTHI RANK 1
For More Join: @IITJEE_Advanced

CHEMISTRY **Max Marks: 60**

SECTION–I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

33. Which of the following pairs of metals liberate NO[Nitric oxide] gas with 20% dil HNO_3

A) Cu, Fe **B)** Sn, Zn **C)** Hg, Pb **D)** Ag, Au

34. The moles of Schottky defects (n_s) and Frenkel defects (n_f) in an ionic crystal is given as: $n_s = N_0.e^{-E/2RT}$ and $n_f = \sqrt{N_o.N_i}.e^{-E/2RT}$ respectively

where N_o is the number of ideal lattice points

N_i is the number of available interstitial sites

E is the activation energy the formation of respective defect.

For NaCl crystal, if the activation energies of formation of Schottky defects and Frenkel defects are 46 and 73.6 kcal/mol, respectively then the fraction of Schoottky and Frenkel defects in a NaCl crystal respectively at 1000 K is

A) 1×10^{-5} and 1.41×10^{-5} B) 1×10^{-5} *and* 1.41×10^{-8}

C) 1×10^{-8} *and* 1.41×10^{-8} D) 1.41×10^{-8} *and* 1×10^{-5}

35. Degree of dissociation (α) of reaction at equilibrium is related to initial vapour density (D) of the reaction as well as at equilibrium vapour density (d) is related as

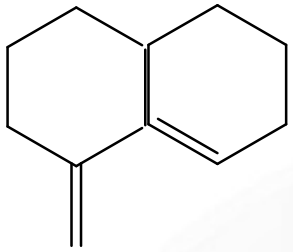
Column 1 (Reaction)		Column II (Degree of Dissociation)	
a.	$2SO_3 \rightleftharpoons 2SO_2 + O_2$	p.	$\frac{D-d}{2d}$
b.	$N_2O_4 \rightleftharpoons 2NO_2$	q.	$\frac{D-d}{d}$
c.	$4PH_3 \rightleftharpoons P_4 + 6H_2$	r.	$\frac{2(D-d)}{d}$
d	$5A \rightleftharpoons 6B + 9C$	s.	$\frac{4(D-d)}{3d}$

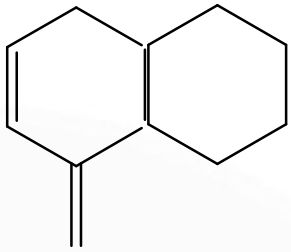
A) $a-r, b-s, c-p, d-q$ **B)** $a-q, b-r, c-p, d-s$

C) $a-q, b-r, c-p, d-s$ **D)** $a-r, b-q, c-s, d-p$

Sri Chaitanya Educational Institutions Infinity Learn


For More Join: @IITJEE_Advanced





36. A) B)

Select true statement(s)

A) Both reaction A & B will give 1,4 addition for major product formation

B) Both reaction A & B will give 1, 2 addition for major product formation

C) A will give 1,2 addition and B will give 1,4 addition for major product formation

D) A will give 1,4 addition and B will give 1,2 addition for major product formation

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.

For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

37. Which of the following statement is/are correct

A) When lead Nitrate is heated to 400^0C only one paramagnetic gas is released

B) Complete hydrolysis of PCl_5 gives H_3PO_4 *and* HCl

C) $N_2O_{5(g)} + 2NaOH_{(aq)} \rightarrow 2NaNO_3 + H_2O$

D) CCl_4 can be hydrolyzed but hydrolysis of $SiCl_4$ can form $H_4SiO_4 + HCl$

38. Which of the following statements is/are correct

A) The chemical process in the production of steel from hematite ore involves reduction followed by oxidation

B) The process of cupellation is used to remove Ag impurity from Pb metal

C) Below 710^0C, CO is more effective reducing agent than carbon

D) Poling process is used to remove SnO impurity from impure Sn

Sri Chaitanya Educational Institutions
Infinity Learn by Sri Chaitanya
41 Years of Excellence 1986-2026
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 RANK 1 SINGARAJU VENKAT KOUNDINYA sri chaitanya 6th-12th Class
JEE Advanced 2023 RANK 1 VAVILALA CHIDVILAS REDDY sri chaitanya 6th-12th Class
NEET 2023 RANK 1 BORA VARUN CHAKRAVARTHI sri chaitanya 6th-12th Class
For More Join: @IITJEE_Advanced

39. A solution of 0.2 g of a compound containing Cu^{2+} *and* $C_2O_4^{2-}$ ions on titration with 0.02 M $KMno_4$ in presence of H_2SO_4 consumes 22.6 mL oxidant. The resulting solution is neutralized by Na_2CO_3, acidified with dilute acetic acid and titrated with excess of KI. The liberated iodine required 11.3 mL of 0.05 M $Na_2S_2O_3$ for complete reduction based on the given data identify correct statements among the following .

A) milli moles of Cu^{2+} ions in the solution = 0.565

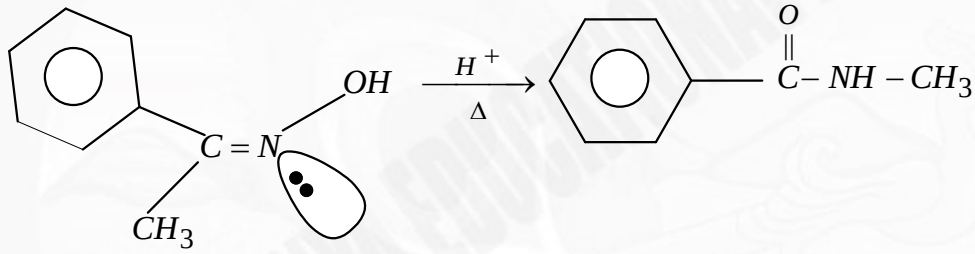
B) milli moles of $C_2O_4^{2-}$ ions in the solution = 1.13

C) molar ratio of $C_2O_4^{2-}$ to Cu^{2+} is 2

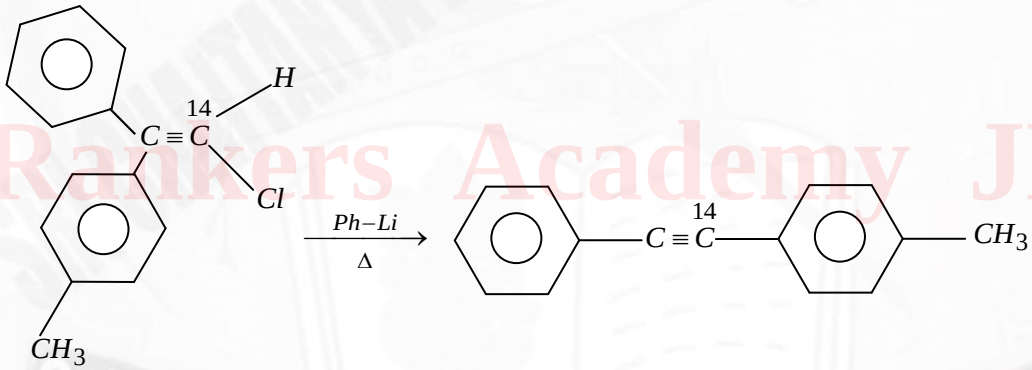
D) Cu^{2+} ions with KI reduced to Cu

40. Which of the following reaction is not representing correct major product

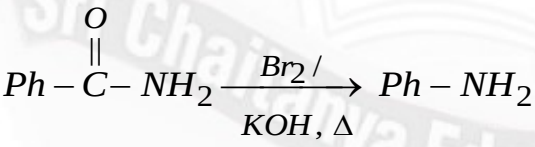
A)



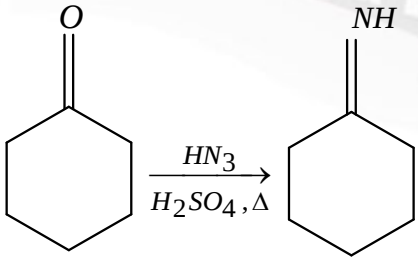
B)



C)



D)



Sri Chaitanya Educational Institutions
Infinity Learn
41
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 SINGARAJU VENKAT KOUNDINYA RANK 1
JEE Advanced 2023 VAVILALA CHIDVILAS REDDY RANK 1
NEET 2023 BORA VARUN CHAKRAVARTHI RANK 1
For More Join: @IITJEE_Advanced

SECTION-III
(NUMERICAL VALUE)

- This section contains **EIGHT (08)** questions stems.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks: +4*** If ONLY the correct numerical value is entered at the designated place;
- **Zero Marks:0** in all other cases

41. Sum of oxidation state (magnitude) and coordination number of cobalt in $Na\left[Co(bipy)Cl_4\right]$

42. Number of electrons present in 4f orbital of Ho^{+3} ion (Z=67) is

43. Count the number of correct statements.

A) Minimum potential required for electrophoresis is called zeta potential

B) Tyndall effect increases with decrease in size of colloidal particle and increase in viscosity of dispersion medium

C) Minimum amount of electrolyte in milli moles required to Coagulate 100 ml of sol in two hours is known as coagulating value.

D) Both physical and chemical adsorption decreases with increase of temperature

E) Emulsions does not show Brownian motion and Tyndall effect.

F) Zeolites are shape-selective catalysts due to their specific pore structure

G) Delta at the mouth of a river is formed due to the coagulation of the colloidal particles of clay in sea water by the electrolytes present in the river water

H) Artificial rain is caused by the coagulation of colloidal water droplets in the clouds by an electrolyte/electrified sand

I) Cotrelle precipitator is used in the precipitation of the colloidal waste from the smoke in industrial chimney's

J) A colloid formed by adding small amount of KI to $AgNO_3$ solution is negatively charged

K) Micelles are formed only above the Kraft temperature (T_k) and below critical micellisation concentration

L) Mixing of hydrated ferric oxide sol and aresnious sulphide cause mutual coaulation due to the neutralization of charges on the particles

M) An water in oil emulsion can be diluted by addition of water

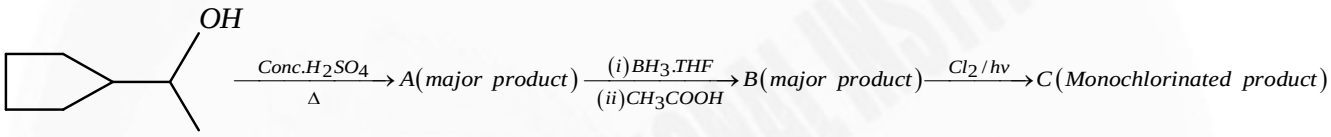
Sri Chaitanya Educational Institutions
Infinity Learn
41


For More Join: @IITJEE_Advanced

44. In a constant volume calorimeter, 3.5 g of CO gas was burnt in excess of oxygen at 298.0K. The temperature of the calorimeter was found to increase from 298.0 K to 298.45 K due to the combustion process. Given that the heat capacity of the calorimeter is 2.5 $kJ\ K^{-1}$, the numerical value for the enthalpy of combustion of the gas $kJ\ mol^{-1}$ in nearest integer is [Use R = $R = 8 \times 10^{-3}\ KJ\ mol^{-1}\ K^{-1}$]

45. A chemical reaction is completed in 6 hours at $30^0 C$ the temperature coefficient (Q_{10}) of the reaction is 3. If the reaction is carried out in 1 hour, approximately at what temperature in kelvin must the reaction be carried out. $(\ln 6 = 1.8, \ln 3 = 1.1)$

46.

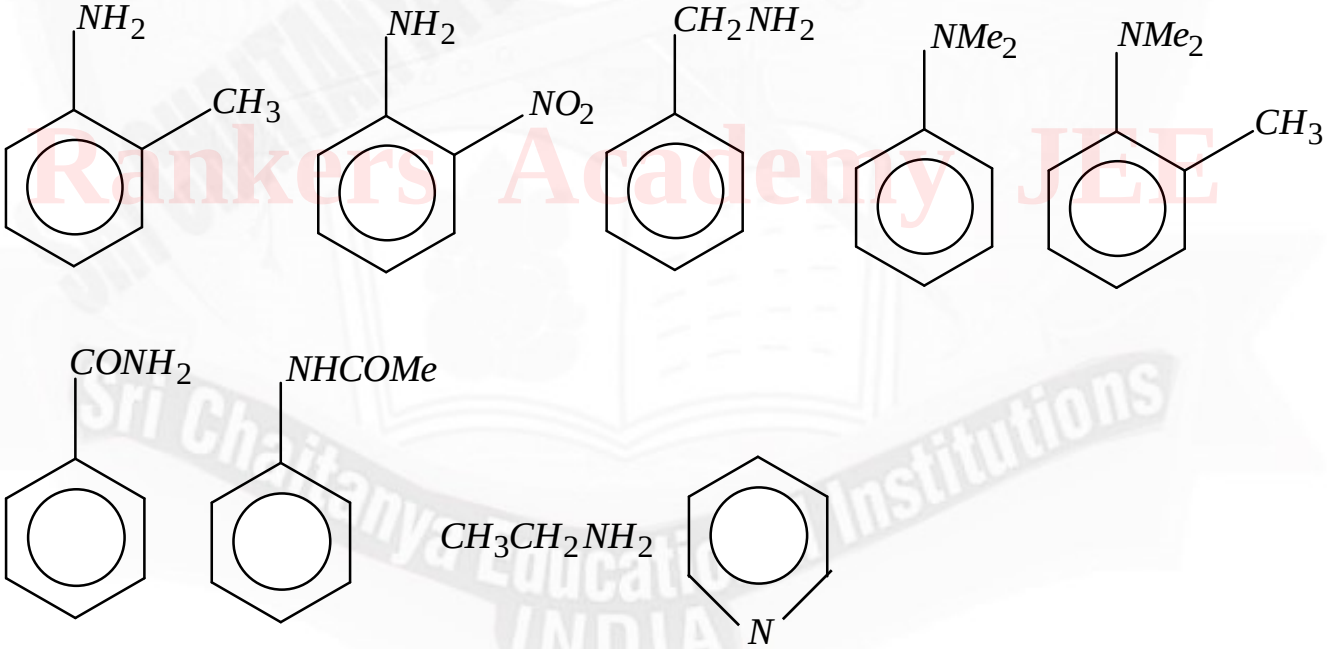


Number of fractions obtained during fractional distillation of C:

47. $A\ (C_4H_5Br) \xrightarrow{Mg/Et_2O} \xrightarrow{D_2O} CH_3CH_2C \equiv CD$

Total number of possible A is/are

48. How many of the following compounds is/are more basic than aniline?



Sri Chaitanya Educational Institutions
Infinity Learn
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 SINGARAJU VENKAT KOUNDINYA RANK 1
JEE Advanced 2023 VAVILALA CHIDVILAS REDDY RANK 1
NEET 2023 BORA VARUN CHAKRAVARTHI RANK 1
For More Join: @IITJEE_Advanced

Sri Chaitanya Educational Institutions | Infinity Learn

41 Years of Excellence 1986-2026

LEADING BY MILES SRI CHAITANYA DOMINATES

JEE ADVANCED 2025

29 Ranks in Top 100 in All-India Open Category



4 Students in Top 11 in JEE-Advanced 2025, All India Open Category

Ranchers Academy JEE

16 RANK DEVDUTTA MAJHI HT. No. 255053116*	18 RANK DHARMANA GNANA RUTVIK SAI HT. No. 256055278	19 RANK VANGALA AJAY REDDY HT. No. 256131009	23 RANK AKSH GOGI HT. No. 252071075*	26 RANK P HEMA SAI SURYA KARTHIK HT. No. 256033006	27 RANK SARKARSAMUDRA HT. No. 252071105*
30 RANK OM PRAKASH BEHERA HT. No. 252021018*	32 RANK SUNKARA SAI RISHANTH REDDY HT. No. 256165327	34 RANK DHRUBA JYOTHI PANJA HT. No. 252048248*	35 RANK BHAVESH JAYANTHI HT. No. 251043080	36 RANK ADVAY MAYANK HT. No. 252104113*	37 RANK KARMANYA GUPTA HT. No. 252081477*
42 RANK MD ANAS HT. No. 252046210*	45 RANK RAMIT GOYAL HT. No. 257001113*	52 RANK MAULIK JAIN HT. No. 252079407*	54 RANK GARV HT. No. 252056188*	59 RANK LARISSA HT. No. 252079071*	60 RANK ARYAN BALABADRULA HT. No. 256132077
63 RANK SAMYAJYOTI BISWAS HT. No. 255058456*	64 RANK AARUSH ANAND HT. No. 251006176*	72 RANK RUSHMITH BANDARI HT. No. 256168048	78 RANK KORIKANA RASAGNYA HT. No. 256057046	87 RANK LAKSHYA SHARMA HT. No. 252070079*	91 RANK AVANEESH BANSAL HT. No. 251113130*	95 RANK KAVYA AGGARWAL HT. No. 252079121*

BELOW 100 ALL INDIA OPEN CATEGORY RANKS: 29	BELOW 500 ALL INDIA OPEN CATEGORY RANKS: 113	BELOW 1000 ALL INDIA OPEN CATEGORY RANKS: 205	BELOW 1000 ALL INDIA CATEGORY RANKS COUNT: 745	NUMBER OF QUALIFIED RANKS: 4,212

0.10 66 15 15 15 www.srichaitanya.net

For More Join: @IITJEE_Advanced